MARKDOWN · MILESTONES_COFFEE_SIMPLIFIED_PROJECT_PLAN.MD

# Milestones Coffee - Simplified Project Plan

## English-Only QR Code Menu Platform
## (V1 - No Arabic, No Payments)

**Version:** 1.0
**Date:** October 3, 2026
**Status:** Planning Phase
**Scope:** Core digital menu system with QR routing and branch management

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## 1. Executive Summary

Milestones Coffee is transitioning from static printed menus to a digital QR code-based menu system. This simplified version focuses on:

- **Customer Experience:** Scan QR code → Auto-detect branch → Browse English menu → View products → Show at counter
- **Staff Experience:** Branch managers update availability in real-time
- **Business Goal:** Replace static menus, maintain centralized catalog, enable fast availability updates
- **Payment:** NOT included - customers order at counter and pay via existing POS system

**Key Difference from Full Plan:** English-only interface, no payment/checkout flow, no customer accounts/authentication.

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## 2. Objectives & Success Measures

### 2.1 Core Objectives

1. Replace static printed menus with digital QR-based access
2. Maintain single centralized product catalog across all branches
3. Enable branch managers to control staff availability and product availability
4. Preserve permanent QR links (never expire or change)
5. Provide analytics on menu browsing without requiring customer orders

### 2.2 Success Metrics

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- QR code correctly resolves branch location ≥99% of the time
- Staff availability updates visible within 10 seconds
- Menu loads on mobile in <2 seconds
- Supports 375px+ mobile widths (all phones)
- Analytics track browsing patterns (no order data required)

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## 3. Scope - What's Included (V1)

### 3.1 Customer-Facing App (Menu Viewer)

**Platform:** Mobile web app (responsive, no native app)

**Core Features:**

- QR code scanning entry point
- Automatic branch detection (QR encodes branch code)
- Browse menu by category
- Search across all products
- Product detail view with:
  - Photo/image
  - Description
  - Price (branch-specific)
  - Variants (size: Small/Medium/Large)
  - Options/Add-ons (temperature, milk type, flavor)
  - Allergens & dietary attributes
  - Availability status
- Labels: New, Popular, Sold Out, Seasonal
- Counter summary (what you've selected) with indicative total
- Filters: Hot/Cold, Coffee/Non-Coffee, Vegan, Vegetarian, Dairy-Free, Gluten-Free, Sugar-Free, New, Popular, Seasonal
- Simple feedback option (optional stars or comment)

**UX Requirements:**

- Mobile-first design (375px, 390px, 430px widths)
- Large touch targets (min 48px)
- Category navigation always visible

- Readable prices & descriptions
- Branch name & language visible at top
- Restrained animations
- Persistent language/branch during session

**Technology:**

- React web app
- Tailwind CSS for responsive design
- PWA capabilities (installable, offline menu cache)

### 3.2 Management Dashboard (Admin Portal)

**User Types:** Admin, Branch Manager, Viewer

**Branch Manager Features:**

- View assigned branch details:
  - Branch name, address, contact
  - Operating hours & holiday overrides
  - Timezone
  - Branch-specific images/signage
- Update availability in real-time:
  - Toggle products: Available / Sold Out
  - Fast toggle UI (ideally icon button, not dropdown)
  - Timestamp showing when last updated
  - Optimistic UI (updates feel instant)
- View which staff are assigned to branch
- Search/filter products for quick availability updates

**Admin Features:**

- Manage all branches:
  - Create, edit, delete branches
  - Set timezone, hours, holiday dates
  - Assign managers & staff to branches
- Manage master catalog:
  - Create, edit, delete products
  - Define categories & subcategories
  - Upload product images

    - Set base prices
    - Define variants (Size: S/M/L) & options (Temp, Milk, Flavor)
    - Tag products (New, Popular, Seasonal)
    - Set dietary attributes (Vegan, Vegetarian, Dairy-Free, Gluten-Free, Sugar-Free)
    - List allergens
    - Priority/sort order
- Branch pricing override:
    - Set branch-specific prices for any product
    - Default to master price if not overridden
- QR code generation & management:
    - Generate unique QR codes per branch
    - Display QR codes for printing/signage
    - Track QR code deployment
- Analytics dashboard:
    - Menu views per branch (daily/weekly/monthly)
    - Most viewed products
    - Search queries
    - Product click-through rates
    - Feedback submissions
- Staff management:
    - Create user accounts
    - Assign roles (Admin, Branch Manager, Staff, Viewer)
    - Permissions control per role

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## 4. Scope - What's EXCLUDED (V1)

**Explicitly NOT included in this version:**

- Customer accounts & authentication
- Shopping cart functionality
- Checkout flow
- Payment processing (any payment gateway)
- Order history or order management
- Kitchen display systems (KDS)
- Delivery/shipping management
- Table ordering

- Loyalty programs or rewards
- POS system integration
- SMS/email notifications
- Inventory management (only availability on/off)
- Arabic language or RTL layout
- Multi-language support beyond English

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## 5. Database Design

### 5.1 Core Entities

#### Branch

```
- id (UUID, primary key)
- code (unique, e.g., "DIFC", "MARINA")
- name (e.g., "Milestones DIFC")
- description
- address
- phone
- timezone (e.g., "Asia/Dubai")
- latitude / longitude (for map)
- operating_hours (JSON: {day: {open: "06:00", close: "22:00"}})
- holiday_dates (date array)
- cover_image_url
- logo_image_url
- status (active, inactive, coming_soon)
- created_at / updated_at
```

#### Product (Master Catalog)

```
- id (UUID, primary key)
- sku (unique identifier, e.g., "CAFF-ESP-001")
- name (e.g., "Espresso")
- description (detailed description)
- category_id (foreign key to Category)
- subcategory_id (foreign key to SubCategory)
- base_price (AED)
- image_url
- status (active, inactive, archived)
- display_order (priority for sorting)
- is_new (boolean, 30-day flag)
- is_popular (boolean, manual flag)
- is_seasonal (boolean)
- nutritional_info (JSON: {calories, protein, fat, carbs, sugar})
```

```
- nutritional_info (JSON: {calories, protein, fat, carbs, sugar})
- dietary_tags (JSON array: ["vegan", "vegetarian", "dairy-free", "gluten-free
- allergens (JSON array: ["nuts", "dairy", "gluten"])
- created_at / updated_at
```

### Product Variant (e.g., Size)

```
- id (UUID)
- product_id (FK to Product)
- name (e.g., "Size")
- display_order
- options (JSON array of variant options with names & relative price adjustmen
  - {name: "Small", price_adjustment: -0.50}
  - {name: "Medium", price_adjustment: 0}
  - {name: "Large", price_adjustment: +0.50}
```

### Option Group (e.g., Temperature, Milk Type)

```
- id (UUID)
- product_id (FK to Product)
- name (e.g., "Temperature", "Milk Type")
- display_order
- options (JSON array)
  - {name: "Hot", price_adjustment: 0}
  - {name: "Iced", price_adjustment: 0}
```

### Category

```
- id (UUID)
- name (e.g., "Coffee", "Beverages")
- description
- display_order
- icon_url
- is_active (boolean)
```

### Branch Pricing Override

```
- id (UUID)
- branch_id (FK to Branch)
- product_id (FK to Product)
- override_price (AED)
- created_at / updated_at
```

### Product Availability (Real-time State)

```
- id (UUID)
- branch_id (FK to Branch)
- product_id (FK to Product)
- is_available (boolean)
- last_updated_at (timestamp)
- updated_by_user_id (FK to User)
```

### User

```
- id (UUID)
- email (unique)
- password_hash
- role (admin, branch_manager, staff, viewer)
- assigned_branch_id (FK to Branch, nullable)
- first_name
- last_name
- phone
- status (active, inactive)
- created_at / updated_at
```

### Analytics Event

```
- id (UUID)
- event_type (page_view, product_view, search, feedback)
- branch_id (FK to Branch)
- product_id (FK to Product, nullable)
- search_query (nullable)
- feedback_text (nullable)
- feedback_rating (nullable)
- user_agent (device info)
- created_at
```

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## 6. API Endpoints

### 6.1 Customer Menu Endpoints (Public)

#### GET /api/branches/:branch_code

Resolve branch from QR code

```
{
```

```
  "id": "uuid",
  "code": "DIFC",
  "name": "Milestones DIFC",
  "timezone": "Asia/Dubai",
  "operating_hours": {...},
  "is_open": true,
  "cover_image_url": "https://..."
}
```

### GET /api/branches/:branch_id/menu

Get full menu for a branch (products + availability + branch-specific pricing)

```
{
  "categories": [
    {
      "id": "uuid",
      "name": "Coffee",
      "products": [
        {
          "id": "uuid",
          "name": "Espresso",
          "price": 15.00,
          "image_url": "...",
          "is_available": true,
          "variants": [...],
          "options": [...],
          "tags": ["new", "popular"]
        }
      ]
    }
  ]
}
```

### GET /api/products/search

Search products

```
Query params: q (search string), branch_id (for availability), filter (vegan,
Returns: Array of matching products with availability for that branch
```

### POST /api/analytics/events

Log browsing events

```
{
```

```
{
  "event_type": "product_view",
  "branch_id": "uuid",
  "product_id": "uuid"
}
```

#### POST /api/feedback

Submit optional feedback

```
{
  "branch_id": "uuid",
  "rating": 5,
  "comment": "Great menu app!"
}
```

### 6.2 Admin Endpoints (Protected)

#### Branches

- `GET /api/admin/branches` - List all branches
- `GET /api/admin/branches/:id` - Get branch details
- `POST /api/admin/branches` - Create branch
- `PUT /api/admin/branches/:id` - Update branch
- `DELETE /api/admin/branches/:id` - Delete branch
- `GET /api/admin/branches/:id/qr-code` - Generate QR code

#### Products (Master Catalog)

- `GET /api/admin/products` - List all products
- `POST /api/admin/products` - Create product
- `PUT /api/admin/products/:id` - Update product
- `DELETE /api/admin/products/:id` - Delete product
- `GET /api/admin/products/:id/variants` - Get variants
- `POST /api/admin/products/:id/variants` - Add variant
- `PUT /api/admin/products/:id/variants/:variantId` - Update variant

#### Availability (Fast Real-time Updates)

- `GET /api/admin/branches/:branch_id/availability` - Get all availability states
- `PUT /api/admin/branches/:branch_id/products/:product_id/availability` -

Toggle availability (optimistic)

#### Branch Pricing

- `GET /api/admin/branches/:branch_id/pricing` - Get branch price overrides
- `PUT /api/admin/branches/:branch_id/products/:product_id/price` - Set branch price

#### Analytics

- `GET /api/admin/analytics/views` - Menu view statistics
- `GET /api/admin/analytics/products` - Product view statistics
- `GET /api/admin/analytics/searches` - Search query statistics
- `GET /api/admin/analytics/feedback` - Feedback submissions

#### Users & Roles

- `GET /api/admin/users` - List users
- `POST /api/admin/users` - Create user
- `PUT /api/admin/users/:id` - Update user
- `DELETE /api/admin/users/:id` - Deactivate user

### 6.3 Health & System

- `GET /health` - Service health check
- `GET /metrics` - Prometheus metrics (if monitoring enabled)

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## 7. Technology Stack

### Frontend

- **Framework:** React 18+
- **Styling:** Tailwind CSS
- **State Management:** React Query + Context API
- **Build Tool:** Vite
- **PWA:** Workbox for service workers
- **QR Scanning:** html5-qrcode library

### Backend

- **Runtime:** Node.js 18+
- **Framework:** NestJS
- **ORM:** TypeORM
- **Database:** PostgreSQL 15
- **Caching:** Redis 7
- **Authentication:** JWT (no user signup - only staff login)
- **File Storage:** AWS S3 (or local filesystem for dev)
- **Monitoring:** Prometheus metrics + health checks

### Infrastructure

- **Containerization:** Docker (multi-stage builds)
- **Orchestration:** Docker Compose (dev), Kubernetes (production)
- **CI/CD:** GitHub Actions
- **Logging:** Winston (structured logs)

### Development

- **Package Manager:** pnpm
- **Monorepo:** Turbo
- **Code Quality:** ESLint, Prettier
- **Testing:** Jest + React Testing Library

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## 8. Project Architecture

### Monorepo Structure

```
milestones-coffee/
├── packages/
│   ├── api/                # NestJS backend
│   │   ├── src/
│   │   │   ├── modules/
│   │   │   │   ├── branches/
│   │   │   │   ├── products/
│   │   │   │   ├── availability/
│   │   │   │   ├── pricing/
│   │   │   │   ├── analytics/
│   │   │   │   ├── users/
│   │   │   │   └── auth/
│   │   │   ├── common/
```

```
│   │   │   ├── database/
│   │   │   └── main.ts
│   │   └── Dockerfile.production
│   ├── admin-dashboard/     # React admin portal
│   │   ├── src/
│   │   │   ├── pages/
│   │   │   ├── components/
│   │   │   └── App.tsx
│   │   └── Dockerfile.production
│   └── menu-app/            # React customer menu app
│       ├── src/
│       │   ├── pages/
│       │   ├── components/
│       │   └── App.tsx
│       └── Dockerfile.production
├── docker-compose.yml
├── .env.example
├── pnpm-workspace.yaml
└── turbo.json
```

### Database Relationships

```
Branch (1) ──→ (M) Product_Availability
Branch (1) ──→ (M) Branch_Pricing_Override
Product (1) ──→ (M) Product_Availability
Product (1) ──→ (M) Product_Variant
Product (1) ──→ (M) Option_Group
User (M) ──→ (1) Branch
Analytics_Event (M) ──→ (1) Branch
Analytics_Event (M) ──→ (1) Product
```

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## 9. User Roles & Permissions

### Admin

- Full access to all features
- Manage all branches
- Manage master catalog
- Manage all users
- View all analytics

### Branch Manager

- View own branch details

- Update own branch availability (products sold out/available)
- View branch analytics
- View assigned staff

### Staff

- View assigned branch
- View products (read-only)
- Cannot modify anything

### Viewer

- View menu (no admin access)
- Read-only access to products

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## 10. Security & Authentication

- **Admin/Branch Manager Login:** Email + password (JWT token)
- **Token Expiration:** 24 hours
- **Password Hashing:** bcrypt
- **Rate Limiting:** 100 requests/minute per IP
- **CORS:** Restricted to app domains
- **HTTPS:** All production traffic encrypted
- **No Customer Authentication:** Customers access via public QR link (no login needed)

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## 11. Development Phases

### Phase 1: Foundation (Weeks 1-2)

- Set up monorepo (pnpm + Turbo)
- PostgreSQL & Redis setup
- Database schema & migrations
- NestJS API scaffolding
- Authentication module (JWT)
- User & branch management endpoints

### Phase 2: Core Menu System (Weeks 3-4)

- Product catalog API (CRUD)
- Availability management API
- Branch pricing override API
- QR code generation
- Menu viewer app (React)
- Product search & filtering

### Phase 3: Admin Dashboard (Weeks 5-6)

- Admin authentication UI
- Branch management UI
- Product catalog management UI
- Availability toggle UI
- Staff management UI

### Phase 4: Analytics & Polish (Weeks 7-8)

- Analytics event logging
- Analytics dashboard
- Feedback collection
- PWA setup
- Performance optimization
- Responsive design refinement

### Phase 5: Testing & Deployment (Weeks 9-10)

- Unit & integration tests
- Load testing
- Docker containerization
- CI/CD pipeline setup
- Production deployment

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## 12. Key Differences from Full Plan (Arabic + Payment Version)

| FEATURE | FULL PLAN | SIMPLIFIED PLAN |
|---|---|---|

| | | |
|---|---|---|
| Languages | English + Arabic | English only |
| RTL Layout | Yes | No |
| Customer Accounts | Yes | No |
| Shopping Cart | Yes | No |
| Checkout Flow | Yes | No |
| Payment Processing | Yes | No |
| Customer Authentication | Yes | No |
| Order History | Yes | No |
| Loyalty Program | No (mentioned as possible) | No |
| POS Integration | No (planned) | No |
| Kitchen Display | No | No |
| Mobile Widths Supported | All modern | 375px+ all phones |
| Complexity | Medium-High | Low-Medium |
| Estimated Timeline | 12+ weeks | 10 weeks |

## 13. Implementation Priorities

### High Priority (Must Have - V1)

1. QR code branch resolution
2. Menu browsing by category
3. Product details with images & prices
4. Availability toggle for staff (fast UI)
5. Branch-specific pricing
6. Basic analytics

### Medium Priority (Should Have)

1. Product search
2. Filters (vegan, gluten-free, etc.)
3. Branch manager dashboard

4. Admin user management
5. Feedback collection

### Low Priority (Nice to Have - Future)

1. PWA offline support
2. Advanced analytics
3. Product recommendations
4. Seasonal promotions management
5. A/B testing framework

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## 14. Success Criteria for V1 Launch

1. ✅ Menu loads in <2 seconds on 4G mobile
2. ✅ QR code resolves correct branch ≥99% accuracy
3. ✅ Availability updates visible within 10 seconds
4. ✅ Works on all mobile widths (375px+)
5. ✅ Admin can manage 100+ products per branch
6. ✅ Analytics track ≥95% of customer views
7. ✅ Supports up to 20 branches without performance degradation
8. ✅ Zero payment/checkout code in repository
9. ✅ Zero Arabic language assets
10. ✅ Documented API with clear examples

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## 15. Next Steps

1. **Review & Approval** - Stakeholder sign-off on scope
2. **Team Assignment** - Assign developers to frontend/backend
3. **Environment Setup** - PostgreSQL, Redis, Docker, GitHub repo
4. **Sprint Planning** - Break phases into 2-week sprints
5. **Development Kickoff** - Begin Phase 1 (Foundation)

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**Document Status:** Ready for Development
**Last Updated:** October 3, 2026